

## Correlation and Covariance

**Q1. Define Covariance and explain how it differs from Correlation in terms of scale and interpretation.**

Covariance is a statistical measure that shows how two variables change together.

- If both variables increase or decrease together, the covariance is positive.
- If one variable increases while the other decreases, the covariance is negative.
- If there is no relationship, the covariance is close to zero.

Difference between Covariance and Correlation

| Covariance                                       | Correlation                                                    |
|--------------------------------------------------|----------------------------------------------------------------|
| Shows only the direction of the relationship.    | Shows both the direction and the strength of the relationship. |
| Its value depends on the units of the variables. | It is unit-free (does not depend on units).                    |
| It can have any value.                           | Its value always lies between -1 and +1.                       |
| It is difficult to compare different datasets.   | It is easy to compare different datasets.                      |

Example:

If the number of hours studied increases and exam marks also increase:

- Covariance will be positive.
- Correlation will be close to +1, showing a strong positive relationship.

**Q2. What does a positive, negative, and zero covariance indicate about the relationship between two variables?**

- Positive Covariance: Both variables move in the same direction. If one increases, the other also increases.
- Negative Covariance: The variables move in opposite directions. If one increases, the other decreases.
- Zero Covariance: There is no linear relationship between the two variables.

Example:

- Positive: Study hours and marks.
- Negative: Product price and demand.
- Zero: Shoe size and exam marks.

**Q3. Discuss the limitations of covariance as a measure of relationship between two variables. Why is correlation preferred in many cases?**

Limitations of Covariance:

- It depends on the units (scale) of the variables.
- Its value can be very large or very small, so it is difficult to interpret.
- It only shows the direction of the relationship, not its strength.

Why is Correlation Preferred?

- Correlation is unit-free (independent of scale).
- Its value always lies between -1 and +1, making it easy to understand.
- It shows both the direction and the strength of the relationship.
- It allows easy comparison between different datasets.

Conclusion:

Correlation is preferred because it is easier to interpret and compare than covariance.

**Q4. Explain the difference between Pearson's correlation coefficient and Spearman's rank correlation coefficient. When would you prefer to use Spearman's correlation?**

| Pearson's Correlation         | Spearman's Rank Correlation             |
|-------------------------------|-----------------------------------------|
| Uses actual data values.      | Uses ranks of the data.                 |
| Measures linear relationship. | Measures rank (monotonic) relationship. |
| Suitable for continuous data. | Suitable for ranked or ordinal data.    |

Uses of Spearman's Correlation

- When the data is ranked (ordinal).
- When the data is not normally distributed.
- When there are outliers that may affect Pearson's correlation.

Use Pearson's correlation for numerical data with a linear relationship, and Spearman's correlation for ranked or non-normal data.

**Q5. If the correlation coefficient between two variables X and Y is 0.85, interpret this value in context. Can you infer causation from this value? Why or why not?**

A correlation coefficient of 0.85 indicates a strong positive relationship between X and Y. This means that as X increases, Y also tends to increase.

Can we infer causation?

No. A high correlation does not mean that one variable causes the other. The relationship may be influenced by other factors or may simply be a coincidence.

**Q6. Using the dataset below, calculate the covariance between X and Y.**

$$\text{Mean of X} = (2 + 4 + 6 + 8) / 4 = 5$$

$$\text{Mean of Y} = (3 + 7 + 5 + 10) / 4 = 6.25$$

**Calculated  $(X - \bar{X})(Y - \bar{Y})$**

| X | Y  | $X - \bar{X}$ | $Y - \bar{Y}$ | Product |
|---|----|---------------|---------------|---------|
| 2 | 3  | -3            | -3.25         | 9.75    |
| 4 | 7  | -1            | 0.75          | -0.75   |
| 6 | 5  | 1             | -1.25         | -1.25   |
| 8 | 10 | 3             | 3.75          | 11.25   |

$$\text{Calculate } (X - \bar{X})(Y - \bar{Y}) = 9.75 - 0.75 - 1.25 + 11.25 = 19$$

$$\text{Cov}(X, Y) = 19 / (4 - 1) = 19/3 = 6.33$$

Since the covariance is positive, X and Y tend to move in the same direction.

**Q7. Compute the Pearson correlation coefficient between variables A and B:**

means

$$\bar{A} = (10 + 20 + 30 + 40 + 50) / 5 = 30$$

$$\bar{B} = (8 + 14 + 18 + 24 + 28) / 5 = 18.4$$

Step 2: Create the calculation table

| A  | B  | $A - \bar{A}$ | $B - \bar{B}$ | $(A - \bar{A})(B - \bar{B})$ | $(A - \bar{A})^2$ | $(B - \bar{B})^2$ |
|----|----|---------------|---------------|------------------------------|-------------------|-------------------|
| 10 | 8  | -20           | -10.4         | 208                          | 400               | 108.16            |
| 20 | 14 | -10           | -4.4          | 44                           | 100               | 19.36             |
| 30 | 18 | 0             | -0.4          | 0                            | 0                 | 0.16              |
| 40 | 24 | 10            | 5.6           | 56                           | 100               | 31.36             |
| 50 | 28 | 20            | 9.6           | 192                          | 400               | 92.16             |

$$\begin{aligned}\sum(A-A^-)(B-B^-) &= 500 \\ \sum(A-A^-)^2 &= 1000 \\ \sum(B-B^-)^2 &= 251.2\end{aligned}$$

Step 3: Apply Pearson's formula

$$\begin{aligned}r &= \frac{\sum(A-A^-)(B-B^-)}{\sqrt{\sum(A-A^-)^2 \times \sum(B-B^-)^2}} \\ r &= 500 / \sqrt{1000 \times 251.2} = 500 / 501.2 \approx 0.998\end{aligned}$$

Pearson correlation coefficient ( $r$ )  $\approx 0.998$

Interpretation:

There is a very strong positive correlation between A and B. As A increases, B also increases almost perfectly.

**Q8. The following table shows heights (in cm) and weights (in kg) of 5 students. Find the correlation coefficient between Height and Weight.**

Mean of

$$X^- = (150 + 160 + 165 + 170 + 180) / 5 = 165$$

$$Y^- = (50 + 55 + 58 + 62 + 70) / 5 = 59$$

Step 2: Create the calculation table

| X   | Y  | $X - X^-$ | $Y - Y^-$ | $(X - X^-)(Y - Y^-)$ | $(X - X^-)^2$ | $(Y - Y^-)^2$ |
|-----|----|-----------|-----------|----------------------|---------------|---------------|
| 150 | 50 | -15       | -9        | 135                  | 225           | 81            |
| 160 | 55 | -5        | -4        | 20                   | 25            | 16            |
| 165 | 58 | 0         | -1        | 0                    | 0             | 1             |
| 170 | 62 | 5         | 3         | 15                   | 25            | 9             |
| 180 | 70 | 15        | 11        | 165                  | 225           | 121           |

$$\sum(X - X^-)(Y - Y^-) = 335$$

$$\sum(X - X^-)^2 = 500$$

$$\sum(Y - Y^-)^2 = 228$$

Using Pearson's formula

$$r = \frac{\sum(X - X^-)(Y - Y^-)}{\sqrt{\sum(X - X^-)^2 \times \sum(Y - Y^-)^2}}$$

$$r = 335 / \sqrt{500 \times 228} = 335 / 337.64 \approx 0.992$$

Correlation coefficient ( $r$ )  $\approx 0.992$

Interpretation:

There is a very strong positive correlation between Height and Weight. As height increases, weight also tends to increase.

**Q9. Given the dataset , determine whether there is a positive or negative correlation between X and Y. (No need for exact calculation, just reasoning.)**

There is a negative correlation between X and Y.

Reason:

As X increases, Y decreases. This means the two variables move in opposite directions, indicating a negative correlation.

Hence, X and Y have a negative correlation.

**Q10. Two investment portfolios have the following returns (%) over 5 years. Compute the covariance and correlation coefficient, and interpret whether the portfolios move together.**

Mean of X =  $(8 + 10 + 12 + 9 + 11) / 5 = 10$

Mean of Y =  $(6 + 9 + 11 + 8 + 10) / 5 = 8.8$

| X  | Y  | $X - \bar{X}$ | $Y - \bar{Y}$ | Product | $(X - \bar{X})^2$ | $(Y - \bar{Y})^2$ |
|----|----|---------------|---------------|---------|-------------------|-------------------|
| 8  | 6  | -2            | -2.8          | 5.6     | 4                 | 7.84              |
| 10 | 9  | 0             | 0.2           | 0       | 0                 | 0.04              |
| 12 | 11 | 2             | 2.2           | 4.4     | 4                 | 4.84              |
| 9  | 8  | -1            | -0.8          | 0.8     | 1                 | 0.64              |
| 11 | 10 | 1             | 1.2           | 1.2     | 1                 | 1.44              |

$$\sum (X - \bar{X})(Y - \bar{Y}) = 12$$

$$\sum (X - \bar{X})^2 = 10$$

$$\sum (Y - \bar{Y})^2 = 14.8$$

Covariance :

$$\text{Cov}(X, Y) = 12 / 5 - 1 = 12/4 = 3$$

Correlation Coefficient :

$$R = 12 / \sqrt{10 * 14.8} = 12 / 12.17 = 0.986$$

The covariance is positive, and the correlation coefficient is very close to +1. Therefore, the two portfolios have a very strong positive correlation and move together. When the return of Portfolio A increases, the return of Portfolio B also tends to increase.